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Paleomagnetism of some mafic intrusions in the South Carolina Piedmont. I. Magnetic systems with single characteristic directions

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Four mafic and two dioritic igneous bodies intruding along the flanks of the Charlotte belt, within the King's Mountain belt and near the Slate–Kiokee–Charlotte belt boundaries in the South Carolina Piedmont, were studied paleomagnetically. The results suggest that these mafic rocks with a single characteristic magnetization are broadly contemporaneous. A mean paleomagnetic pole position of 38.9°N 120.8°E has been calculated for the six bodies. This pole position falls near the 300 Ma old point of Irving's apparent polar wander path (APWP) for North America. The $\alpha 95$ circle of confidence (10.2°) includes points on the APWP between 250–360 Ma. Anomalously old K–Ar apparent ages, greater than one billion years, are suggestive of excess ^{40}Ar contamination for the mafic Buffalo pluton, whereas apparent ages in the range of 360–395 Ma old are interpreted as a maximum age for the other bodies, due to the possibility of small amounts of excess ^{40}Ar being present. A 10° westerly tilt correction suggested by Dooley and Smith for early Mesozoic diabase brings the mafic pole position of this study to more nearly coincide with the 350 Ma virtual geomagnetic position of Irving's curve, but the test is inconclusive, awaiting better definition of radiometric ages. The simplest interpretation of the data is that the mafic pole position reflects the direction of the geomagnetic field in late Devonian–Carboniferous times. The similarity of this pole position with the points on the APWP for North America provides little evidence for displaced terrains and, with the precision of this pole position, horizontal displacements on the order of that suggested by the Consortium for Continental Reflection Profile (COCORP) results, or by subsequent studies, are not detectable. This and other pole positions from granitic rocks in the southern Appalachian orogen suggest that if displaced terrains exist, the evidence must be found in older rocks, or in other geologic belts.

1. Introduction

Pilot studies over the past several years have indicated an empirical division of the mafic intrusives into those with a single hard magnetic component and those with superposed magnetizations in addition to a viscous component. In this study, the bodies with single characteristic directions of magnetization have been investigated. The units considered are the Buffalo, Dutchman's Creek, Little Creek and Clouds Creek mafic plutons and two minor diorite intrusions. The locations of these units, shown in Fig. 1 are close to the Slate–Kiokee belt margin, or within the King's Mountain belt; whereas the units in which superposed magnetizations occur, characteristically lie in the central part of the Charlotte belt.

Whether or not displaced terrains form as important an element in the structure of the exposed southern Appalachians as the Avalonian terrain does further north (Kent and Opdyke, 1978; Van der Voo et al., 1979; Brown, 1979), is still uncer-

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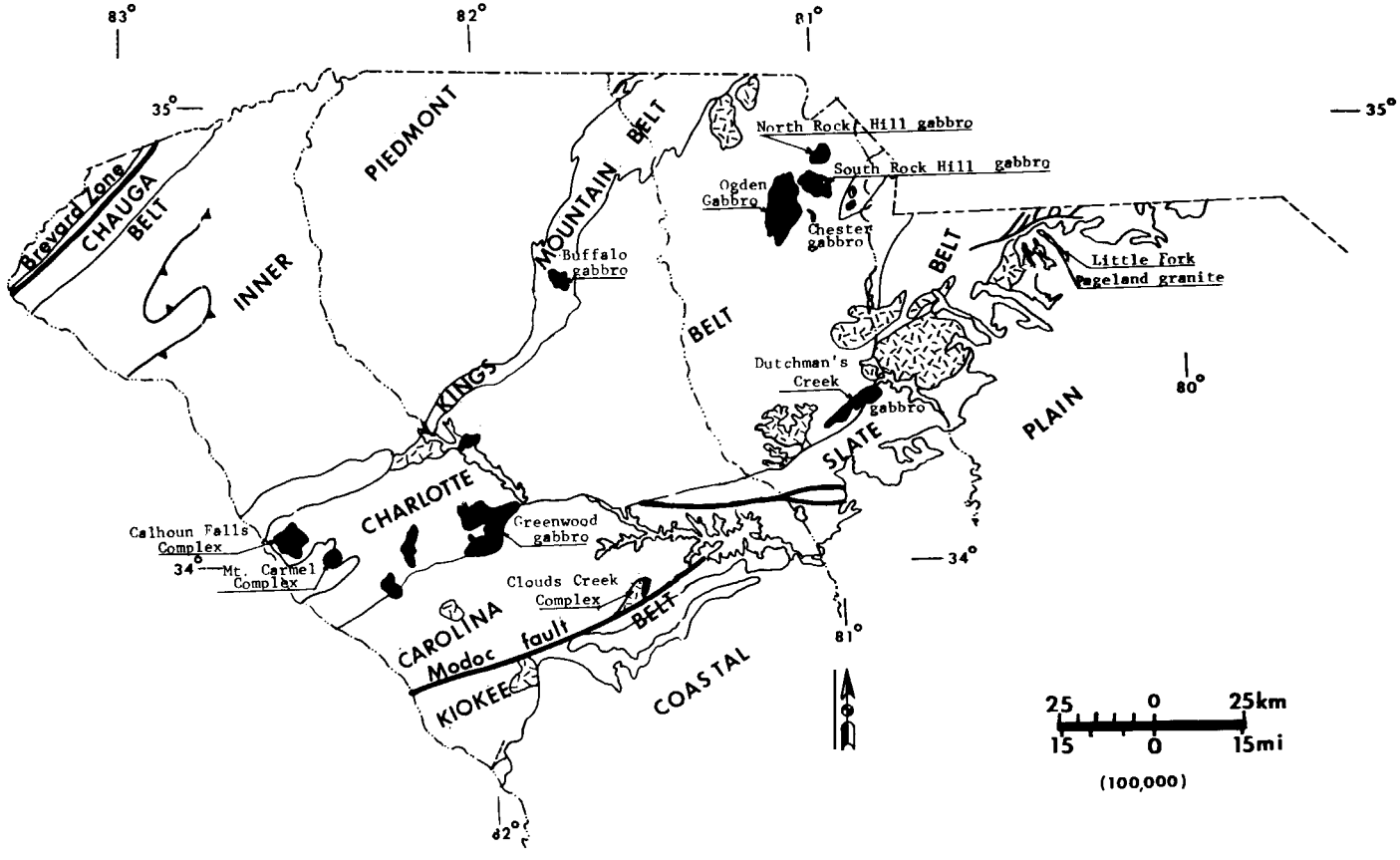


Fig. 1. Generalized geological map of the South Carolina Piedmont (after Secor and Snoke, 1978). Black bodies represent mafic intrusions and hachured areas depict granitic intrusions.

tain, simply because of the few studies so far carried out. Such data as does exist for pre-Mesozoic rocks in the area (Nairn et al., 1978; Watts and Van der Voo, 1979; Edinger et al., 1979; Brown and Barton, 1980; Ellwood and Whitney, 1980; Ellwood et al., 1980; Watts et al., 1980; Ellwood, 1982; Barton and Brown, 1982) provides no clear evidence for any major displacement. The mafic bodies, of which a considerable number exist, provide an obvious target for the first stage of a systematic study to determine if such displaced terrains exist, or to see if rotational tectonic motions may be associated with the horizontal displacement proposed on the basis of the COCORP results (Cook et al., 1979). The degree of horizontal displacement is still in question (Iverson and Smithson, 1982). The gabbros may also be used to test the early Mesozoic tilt correction suggested by Dooley and Smith (1982).

Since several tectonic events are recognized in the history of the southern Appalachians, there is a need for accurate ages of the intrusive units studied, and this magnetic study was combined with conventional whole rock K–Ar studies. The interpretation of the K–Ar apparent ages obtained awaits verification through $^{40}Ar/^{39}Ar$ study, because of the problems of contamination by excess ^{40}Ar found in the course of work on younger mafic rocks in the southern Appalachians (Dooley and Wampler, 1983), and the possibility of argon loss among samples with discordant apparent ages.

1.1. Location

Numerous gabbroic and metagabbroic intrusive complexes, ranging in age from early to late Paleozoic, intrude the Piedmont Province of the southern Appalachian orogenic belt (Overstreet and Bell, 1965a, b; Hermes, 1968; Butler and Ragland, 1969; McSween and Nystrom, 1979). These mafic igneous bodies form an arcuate string from Georgia to Virginia (Butler, 1966; Butler and Ragland, 1969; McSween and Nystrom, 1979). A suite of more than twenty of them intrude the Piedmont of South Carolina. Figure 1 shows only the larger of the bodies in South Carolina. The majority of the South Carolina gabbros intrude almandine–amphibolite facies metamorphic rocks of the Charlotte belt; whereas only a few gabbros intrude metamorphic terrain outside the Charlotte belt along the margins of the Slate–Charlotte belt and the Kings Mountain–Charlotte belt boundaries. The Dutchman's Creek and Greenwood mafic bodies cut across the Slate– Charlotte belt boundary extending into the greenschist regional metamorphic facies rocks of the Slate belt. One pluton, the Buffalo, intrudes the greenschist facies rocks of the King's Mountain belt. Several gabbro stocks cut granitic batholiths intruding the southern margin of the Slate belt in North and South Carolina (Butler and Ragland, 1969). Granites enclosing gabbros, such as the Pageland granite, the Clouds Creek granite and the Lilesville pluton (in North Carolina), are known to have intruded the southern Piedmont around 300 Ma ago (Fullagar, 1971; Fullagar and Butler, 1979; Fullagar and Kish, 1980; Fullagar, 1981).

The larger of the gabbro intrusions in South Carolina include: the Mt. Carmel complex (Medlin, 1968, 1969a; Medlin et al., 1972; and Griffin, 1978), the Greenwood pluton (Hermes, 1968; Medlin et al., 1972; Chalcraft et al., 1981), the Ogden pluton (McSween, 1981), the Rock Hill plutons (Chalcraft, 1970), and the York–Chester complex (Wagener, 1974; Deetz, 1980). The most detailed petrographic and petrologic studies are available for the Chester, the Buffalo, the Dutchman's Creek, and the Mt. Carmel plutons. The concordant to discordant mafic intrusions form a variety of shapes, ranging from oval, to crescent, to dike-like and vary in size from < 1 km^2 to > 60 km^2.

1.2. Petrology

The mafic intrusions vary in composition from basic, hornblende gabbro–pyroxene gabbro–gabbronorite–olivine gabbronorite, representing undifferentiated, irregularly distributed rock varieties within plutons of limited compositional ranges, (e.g., the Ogden and Buffalo plutons) to more differentiated, layered igneous complexes comprised of gabbros intruded by diorites and in some instances syenites (e.g., the Mt. Carmel igneous complex). The plutons are medium- to coarse-grained, porphyritic to non-porphyritic and have a

wide range of textures including poikilitic, ophitic, subophitic and, most commonly, hypidiomorphic-granular. Petrologic data indicate that the bodies have undergone late to post-tectonic emplacement into the mesozone of the Piedmont during the late stages of development of a tectonically active orogenic belt (Buddington, 1959). Many of the bodies impose a partial hornfels contact aureole and/or a garnet–pyroxene skarn upon surrounding metamorphic rocks.

1.3. Geological age

Uncertainty exists in assigning geologic ages for many of the postmetamorphic gabbros in the southern Piedmont. Their generally unmetamorphosed nature and contact aureoles in the surrounding lower Paleozoic metamorphic terrane indicate a late Paleozoic age for the gabbros (McCauley, 1960; Butler and Ragland, 1969). While this stratigraphic and structural evidence can be used to place an older limit for the age of the gabbros intruding the southern Piedmont, it is unfortunately true that the age of sedimentation within the southern Appalachian orogenic belt is poorly known. The only conclusive evidence for the age of sedimentation within the southern Piedmont is that of the rocks of the Slate belt. St. Jean (1973) found an early to late Cambrian trilobite in the Slate belt of North Carolina, and recently an in-situ fossil fauna of trilobites, which include *Paradoxides,* was found within the contact metamorphosed zone of the Slate belt sediments surrounding the Clouds Creek complex in South Carolina (Sara Samson, personal communication, 1982). Thus, the Slate belt metasediments include rocks of an early to late Cambrian age and by inference, the same age has been applied to the original sedimentary material of the Kiokee belt (Secor and Snoke, 1978). Furthermore, geochronologic evidence suggests a Cambrian age for sedimentation in the greater part of the Slate belt (Hills and Butler, 1969; Stromquist and Sundelius, 1969; Fullagar, 1971; Seiders and Wright, 1977; Wright and Seiders, 1977; Black, 1978; Secor and Snoke, 1978; Carpenter et al., 1978).

The area of the southern Piedmont has been subsequently subjected to several different periods of metamorphism. Phases of Taconic, Acadian, and Hercynian metamorphism have been recognized in various portions of the southern Appalachian orogen (Rodgers, 1967, 1970; Fullagar, 1971; Hatcher, 1972; Butler and Fullagar, 1978; Secor and Snoke, 1978; Snoke et al., 1980). Furthermore, the southern orogenic belt has been subsequently uplifted and exhumed differentially, a fact leading to the conclusion that K–Ar and Rb–Sr ages of biotites from metamorphic rocks represent cooling ages (Hadley, 1964; Butler, 1972; Dallmeyer, 1978; Fullagar and Butler, 1979) rather than periods of pervasive orogenic activity (Long et al., 1959; Kulp and Eckleman, 1961; Hurst, 1970). Thus, while penetrative deformation has been documented at 250–300 Ma ago for the Kiokee belt (Secor and Snoke, 1978; Snoke et al., 1980), Hercynian metamorphism has not yet been demonstrated in the rest of the southern Piedmont. The K–Ar and Rb–Sr biotite mineral ages of 250–350 Ma reported for isolated areas of the Slate, Charlotte, King's Mountain and Inner Piedmont belts of the southern Appalachians have been interpreted by Butler (1972) as cooling ages of surfaces subjected to differential uplift, rather than thermal events of a penetratively deformed orogenic event. Furthermore, Butler (1972) suggests that the Taconic was the last well documented orogenic event to pervasively affect the Slate, Charlotte, King's Mountain and Inner Piedmont belts.

Lack of stratigraphic control, plus possible metamorphic effects, make the age of intrusive rocks in the southern Piedmont difficult to establish. The only certain time boundaries that exist are the Cambrian sediments (subsequently metamorphosed), into which igneous bodies have intruded (in the Slate belt), and the late Triassic–early Jurassic age of diabase dikes cutting across metamorphic and igneous structures of the southern Piedmont (Dooley and Smith, 1982; Dooley and Wampler, 1983). Pre-, syn- and post-tectonic bodies can be recognized on the basis of field relations, rock textures and composition. However, closer definition of the geologic age of such bodies rests heavily on radiometric geochronology.

Up to the present time, most of the geochrono-

logical work has concentrated on granitic rocks in the southern Appalachians (Fullagar, 1971; Fullagar et al., 1971; Butler, 1972; Fullagar and Odom, 1973; Jones and Walker, 1973; Odom and Fullagar, 1973; Fullagar and Butler, 1976; Whitney et al., 1976; Butler and Fullagar, 1978; Kish and Fullagar, 1978; Fullagar and Butler, 1979; Fullagar and Kish, 1980; Snoke et al., 1980; Fullagar, 1981; Harper and Fullagar, 1981; Dallmeyer et al., 1981). These studies have established several ages for granitic intrusions, but the most important to this discussion is the 300 Ma age group of post-metamorphic granites in the southern Piedmont. Such granites are clearly post-Taconic and some are intruded by gabbros. Where no such cross cutting relationships have been seen for other gabbros in the southern Piedmont, the age of such basic bodies remains uncertain.

In the course of the present study, conventional K–Ar techniques have been applied to provide preliminary radiometric age constraints for some of the mafic bodies for which age relationships are uncertain (see Table I). Such preliminary results coupled with paleomagnetic data can be used to shed more light on the ages of the post-metamorphic gabbros in the southern Piedmont. With this goal in mind, the following pertinent questions can be posed about the tectonic evolution of the southern Appalachians:

(1) Can the age of post-metamorphic gabbros be better established to compare new paleomagnetic evidence to that for stable cratonic North America?

(2) Can a tilt correction, such as that suggested for the early Mesozoic diabase dikes in the southern Appalachians (Dooley and Smith, 1982) be valid for the post-metamorphic gabbros?

(3) Can paleomagnetic data be used to provide evidence for displaced terrains, to detect rotations about a vertical axis during horizontal displacements as suggested by COCORP data (Cook et al., 1979), or to determine the degree of horizontal displacement, a point now in question (Iverson and Smithson, 1982)?

1.4. Analytical techniques

Samples were cored using a portable field drill equipped with water cooled diamond tipped bits and oriented with a compass and aluminum goniometer. In the laboratory, one or more 22.2 mm (± 0.5) length cylinders were prepared from each core. The natural remanent magnetization (NRM) was measured with a Digico complete results magnetometer interfaced with a Digital MINC II minicomputer.

After NRM direction and intensity measurements, one pilot sample was chosen from each site and progressively demagnetized in alternating fields (AF) at 5 mT increments, from 5 to 50 mT and additionally at 55, 75 and 100 mT increments. A second pilot from each site was progressively demagnetized thermally in 50°C increments from 100 to 650°C.

The stage at which the characteristic magnetization was recovered usually occurred by 15 mT, although in a few instances it was recovered only at fields as high as 35–50 mT. Site mean directions were computed for the different demagnetization stages, and the best representation of the characteristic magnetization was regarded as the demagnetization stage that yielded the highest K, lowest $\alpha 95$ values. These data and the statistical parameters for post-metamorphic gabbros appear in Table I.

2. Results

2.1. The Buffalo gabbroic pluton

The Buffalo gabbroic pluton, located approximately 5 km southwest of Buffalo in Union County, South Carolina, consists of a variety of gabbros, with lesser amounts of norite irregularly distributed within the pluton (Medlin, 1968, 1969a). It has the form of a stock (3×8 km) intruding the King's Mountain belt (Figs. 1, 2). The country rock consists of a biotite–quartz monzonite with a small zone of altered mafic rock along the northern edge of the gabbro (Medlin, 1969b).

Petrologically, its mineralogy consists of olivine, orthopyroxene, clinopyroxene, brown amphibole, anorthoclase and accessory minerals such as biotite, opaques, and reaction rims (Medlin, 1968). The pluton shows affinities with both tholeiitic

TABLE I

Summary of statistical parameters for some paleozoic igneous rocks

Site No.	Dec.°	Inc.°	α95	*a/b*	**R**	*K*	Site		K–Ar app. age (Ma)	Demagnetizing field
							Lat.	Long.		
Buffalo pluton										
1	166.0	−10.8	9.6°	7/7	6.8511	40.3	42.5	−81.8	1,107+22 1,565±30	15 mT
2	166.2	+13.8	8.3	12/12	11.6108	28.2			–	15 mT
3	158.5	+14.5	7.5	5/5	4.9160	104.0			–	25 mT
4	153.9	+16.2	10.9	7/7	6.8110	31.8			–	50 mT
5	159.9	+23.4	12.0	5/7	4.9034	41.4			–	25 mT
6	168.0	+ 6.3	9.1	5/8	4.9448	72.4			–	15 mT
*7	185.6	+30.8	56.5	7/7	4.1372	–			–	15 mT
Average				Sites			Pole Position			
							°N	°E		
	162.2	+10.7	10.8	6/7	5.8726	39.2	39.4	121.5		
Dutchman's Creek pluton										
1	154.1	+24.2	3.3	6/6	5.9780	390.0	34.4	−80.8	396+8; 385±8; 499±10	25 mT
2	146.5	+33.0	3.1	7/7	6.9840	360.0			–	25 mT
3	153.4	+30.0	10.7	6/6	5.8750	11.0				25 mT
*4	180.8	+46.7	8.5	7/7	6.8836	51.6				15 mT
5	161.7	+52.1	10.8	6/7	5.8741	39.7				35 mT
6	164.3	+28.1	8.3	5/5	4.9530	85.2				20 mT
7	164.5	+16.8	5.0	6/6	5.9724	181.3				20 mT
Average				Sites			Pole Position			
							°N	°E		
	160.1	+33.3	11.5	6/7	6.7896	28.5	34.1	122.1		This study
	160.7	+22.6	6.6	11/12	10.7984	49.6	40.4	124.3		Zupan (personal communication, (1982)
Total	161.3	+27.7	6.0	19/19	18.4387	32.1	37.8	122.3		

Jenkinsville diorite plutons										
1	173.1°	+25.4°		–	–	–	–	–	288 ± 6	30 mT
2	148.1	+30.9		–	–	–	–	–	265 ± 6	35 mT
Average	160.9	+28.7								
Clouds Creek complex										
Gabbro										
2	151.7	−3.5	10.3	6/6	5.8852	43.6			–	15 mT
3	144.3	+21.0	9.2	4/4	3.9704	101.4			–	15 mT
4	156.8	+11.9	2.6	7/7	6.9889	540.8			–	20 mT
*5										
6	160.3	−2.4	26.1	6/7	5.3380	7.6			–	20 mT
Average				Sites			Pole Position			
							°N Lat.	°E Long.		
	153.4	6.8	15.6	4/5	3.9155	35.5	45.5	137.9		
Granite										
1	339.3	19.7	31.2	6/10	5.0991	5.6	59.4	144.4		
Pageland complex										
Little Fork gabbro										
1	181.0	42.7	5.1	6/6	5.9709	172.0			293 ± 6	15 mT
2	172.7	37.7	7.2	6/6	5.9429	87.5			285 ± 6	15 mT
Average				Sites			Pole Position			
							°N Lat.	°E Long.		
	176.7	40.3	17.7	2/2	1.9950	201.9	32.1	103.2		
Mean for all intrusive bodies										
	162.3	23.9	10.17	5/5	4.930	26.73	38.9	120.8		

$\alpha 95$, the semi-angle of the cone of confidence; a/b indicates the number of samples used in the computation of the mean (a) and the total number of cores tested (b). **R** is the length of the mean vector, K is Fisher's (1953) precision parameter. The demagnetizing field reported is that at which the characteristic magnetization was obtained. K–Ar ages were calculated using Steiger and Jaeger (1977) decay constants: $n\,^{40}K/n\,K = 0.0001167$; $\lambda_\epsilon = 0.581 \times 10^{-10}\ y^{-1}$; $\lambda_\beta = 4.9162 \times 10^{-10}\ y^{-1}$. All the K–Ar apparent ages presented are new determinations.

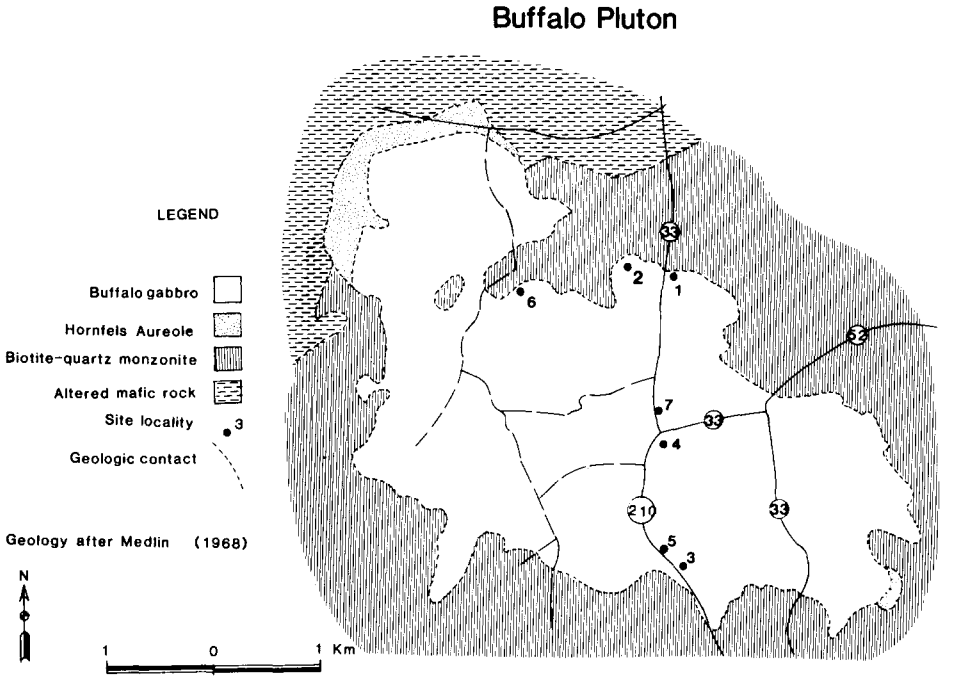


Fig. 2. Generalized geological map of the Buffalo pluton with sample locations (after Medlin, 1968).

and calc-alkaline basaltic compositions.

Seven sites were sampled with an average of six to eight specimens per site for paleomagnetic study (Fig. 2). The attempt to distribute sites over the whole pluton is limited by the number and the accessibility of acceptable in-situ outcrops. Medlin (1968) produced detailed outcrop-boulder locations for his geologic map of the Buffalo pluton; over twenty such sites were visited and only seven selected for paleomagnetic study.

The NRM intensities range between 9.5×10^{-2} A m^{-1} and 6.0×10^{-3} A m^{-1}, averaging 2.2×10^{-2} A m^{-1}. Normalized demagnetization curves (Fig. 3) show considerable variation in the rate of decay of the remanent intensity. The characteristic magnetization appears typically when 20–80% of the NRM intensity has been removed in the fields of 15–25 mT; occasional cores show much harder magnetization and, at 100 mT, as much as 70% of the NRM remains (sample DD18-17-4, Fig. 3). Such samples are not common and do not appear to be site specific. A small number of cores had to be rejected, either because no stable end point was reached, or the direction of magnetization differed by more than 40° from the mean.

The behavior of most cores during cleaning in alternating and thermal fields is typical of basaltic rocks whose major magnetic carrier is titanomagnetite (De Boer, 1968; Smith and Noltimier, 1979; and De Boer and Snider, 1979) which include a mixture of a single and multiple domain materials (Stacey and Bannerjee, 1974). In a number of cases, thermal demagnetization indicates that the principal carrier of remanence is hematite (Fig. 3b, sample DD18-17-1). As is apparent from the normalized stable end point curves (Fig. 4), the remanence directions change very little during cleaning. During thermal demagnetization, fluctuating intensity values were found and the data are plotted as stereograms. These are, however, samples where the NRM directions are dispersed. Figure 5 contrasts the NRM directions of four sites with their cleaned directions. A relatively tight cluster of site mean directions is seen in Fig. 6.

The site mean directions of magnetization and standard Fisher (1953) statistics are shown for the Buffalo pluton in Fig. 6. This declination and in-

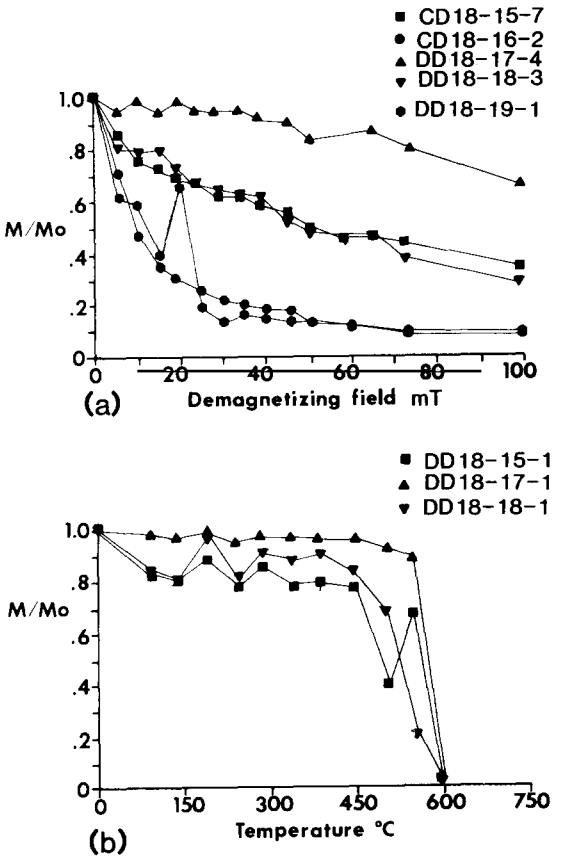


Fig. 3. Alternating frequency (a) and thermal (b) demagnetization spectra for the Buffalo pluton. Site 1 = CDU18-15-7; site 2 = CDU18-16-2; site 3 = DD18-17-4; site 4 = DD18-18-3.

clination corresponds to a pole position of 39.4°N 121.5°E for the Buffalo pluton. Applying a 10° tilt correction about an Appalachian trend (Dooley and Smith, 1982) yields a declination of 165° and an inclination of 1° for a westward tilting of the Appalachian surface, corresponding to a pole position of 14.9°N 119.7°E, and a declination of 152° and inclination of 20° for an eastward tilting of the Appalachian surface, corresponding to a pole position of 31.3°N 130.6°E (Table II).

A K–Ar phlogopite age (Long et al., 1959) of 309 ± 11 Ma for metamorphic rocks of the King's Mountain belt has been interpreted by Hadley (1964) as a cooling age. Medlin (1968) suggested that the Buffalo complex was younger than the 309 ± 11 Ma mineral age. Preliminary conventional K–Ar ages were determined in duplicate for two samples taken approximately 2 m apart at site 2 in the Buffalo pluton and yield whole rock apparent ages of 1,107 ± 22 Ma and 1,565 ± 30 Ma (Table I). Analytical techniques for these analyses are described in detail in Dooley and Wampler (1983) and thus will not be described here. These ages reflect a problem with excess ^{40}Ar contamination, since they are very old relative to the country rocks in which they intrude. The presence of excess ^{40}Ar contamination has been documented for other mafic rocks of tholeiitic composition in the southern Appalachians and other areas of the world (Dooley and Wampler, 1983; Sutter and Smith, 1979; Dalrymple et al., 1975). The significance of conventional K–Ar determinations of gabbros in South Carolina is discussed in another paper (Dooley, Manzoni and Nairn, 1983).

2.2. The Dutchman's Creek gabbro

The Dutchman's Creek gabbro pluton consists of a progressively differentiated suite of rocks including olivine, pyroxene, hornblende–pyroxene and anorthositic gabbros (McSween and Nystrom, 1979). This elongate pluton occurs in Fairfield County, South Carolina within a synclinorium of almandine–amphibolite facies metasedimentary and metavolcanic rocks of the Charlotte belt, and its southern edge is flanked by portions of the Slate belt greenschist metamorphic rocks (Fig. 7, McSween and Nystrom, 1979). The gabbro has ophitic to subophitic textures along its margins and hypidiomorphic granular texture within its center, indicating a relatively simple and uninterrupted cooling history, with differentiation only occurring between its margin and center. Differentiation patterns within the Dutchman's Creek gabbro are similar to those of other layered complexes, even though the degree of differentiation is less (Nockolds, 1954; Taylor and Forester, 1973).

The mineralogy of the gabbro body varies and includes the minerals plagioclase, clinopyroxene, orthopyroxene, amphibole and biotite. Opaques occur as anhedral crystals, or as large pools partially or completely enclosing plagioclase (McSween, 1972). Olivine occurs as an accessory mineral and sometimes is surrounded by idding-

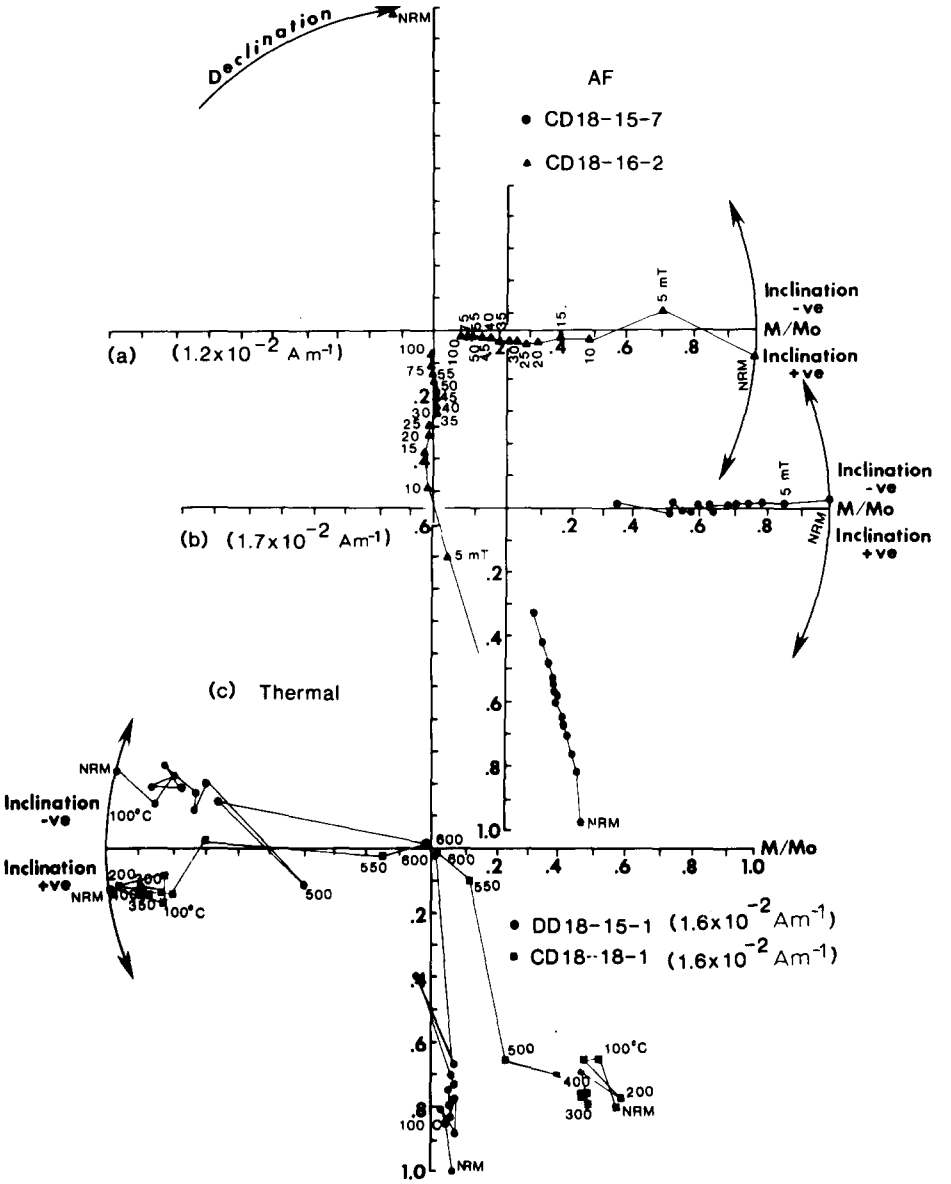


Fig. 4. Stable end point curves for alternating frequency (a, b) and thermal (c) demagnetization of Buffalo pilot samples. NRM intensity values are given in A m^{-1} (in parentheses).

site. Altered olivine crystals are enclosed by amphiboles, clinopyroxenes and opaque phases. Early crystallization of Fe–Ti oxides occurs, but McSween and Nystrom (1979) suggest that many of the grains have subsequently undergone subsolidus oxidation of ilmenite to hematite.

Seven sites were sampled along the elongate axis of the Dutchman's Creek gabbro (Fig. 7). AF and thermal demagnetization spectra, characteristic of the typical behavior of the Dutchman's Creek gabbro, are shown (Fig. 8) and stable end point curves are given (Fig. 9).

AF demagnetization spectra yield characteristic magnetizations at 25 mT, a slightly higher coercive

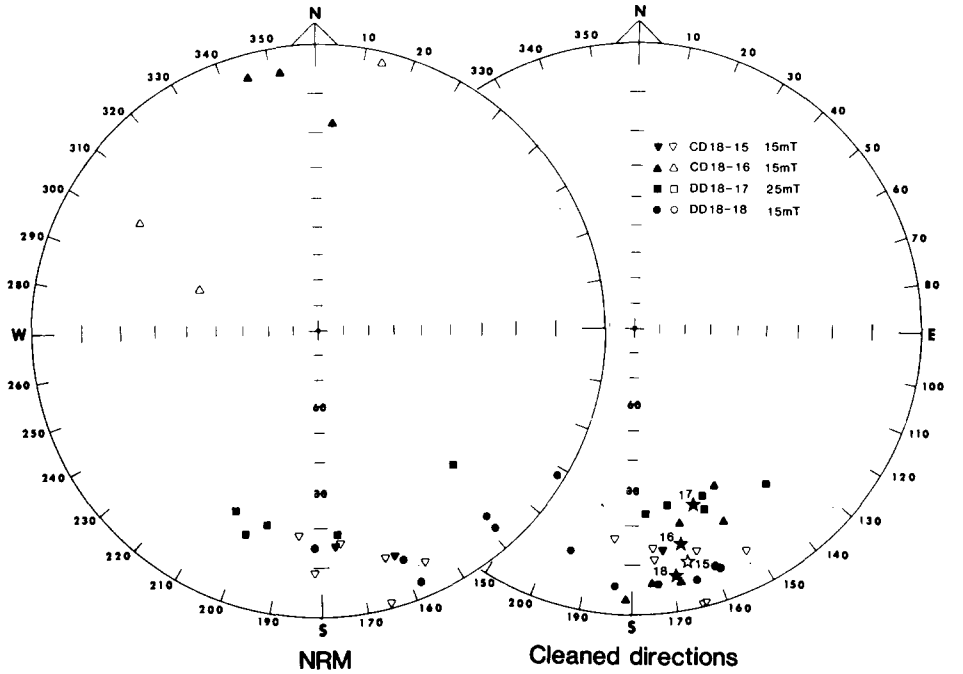


Fig. 5. NRM directions contrasted with the best cleaned directions for the Buffalo pluton. The stars represent site mean directions for four sites: site 1 = CD18-15; site 2 = CD18-16; site 3 = DD18-17; site 4 = DD18-18. Solid figures are lower (positive) and open are upper (negative) hemisphere projections.

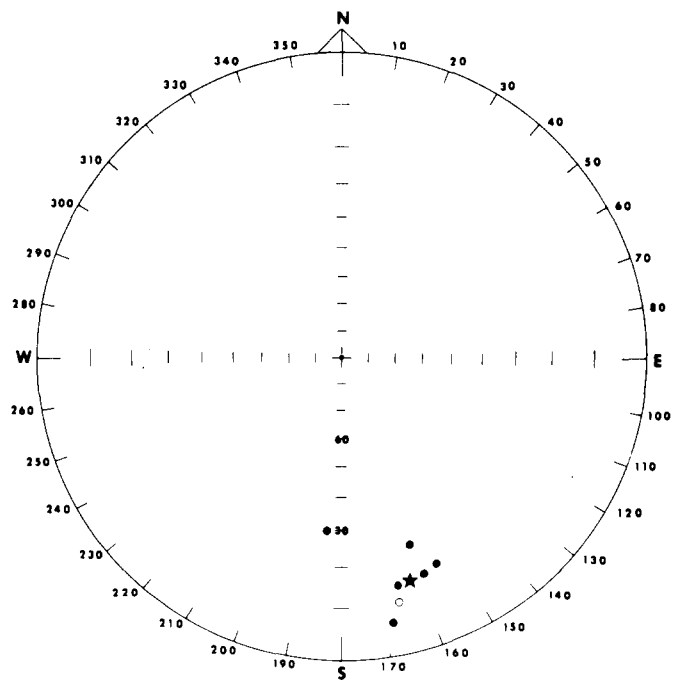


Fig. 6. Site mean directions for seven sites of the Buffalo pluton.

TABLE II

Tilt corrections about the Appalachian trend

Gabbro	Uncorrected data				10° East tilt corrected				10° West tilt corrected			
	Dec.	Inc.	Pole Position		Dec.°	Inc.°	Pole Position		Dec.°	Inc.°	Pole Position	
			°N	°E			°N	°E			°N	°E
Buffalo	162.2	+10.7	39.4	121.5	152.0	20.0	31.3	130.6	165.0	1.0	14.9	119.7
Dutchman's Creek	161.3	+27.7	37.8	122.3	140.0	37.0	23.1	140.0	155.0	17.0	40.9	132.7
Clouds Creek	153.4	+6.8	45.4	137.9	148.0	17.0	38.0	140.1	152.0	−3.0	48.6	143.7
Little Fork	176.7	+40.3	32.1	103.2	130.0	50.0	9.3	141.4	150.0	30.0	31.7	133.9
Jenkinsville Diorites	160.9	+28.7	37.2	122.0	157.0	37.0	30.9	123.9	141.0	17.0	33.4	146.9

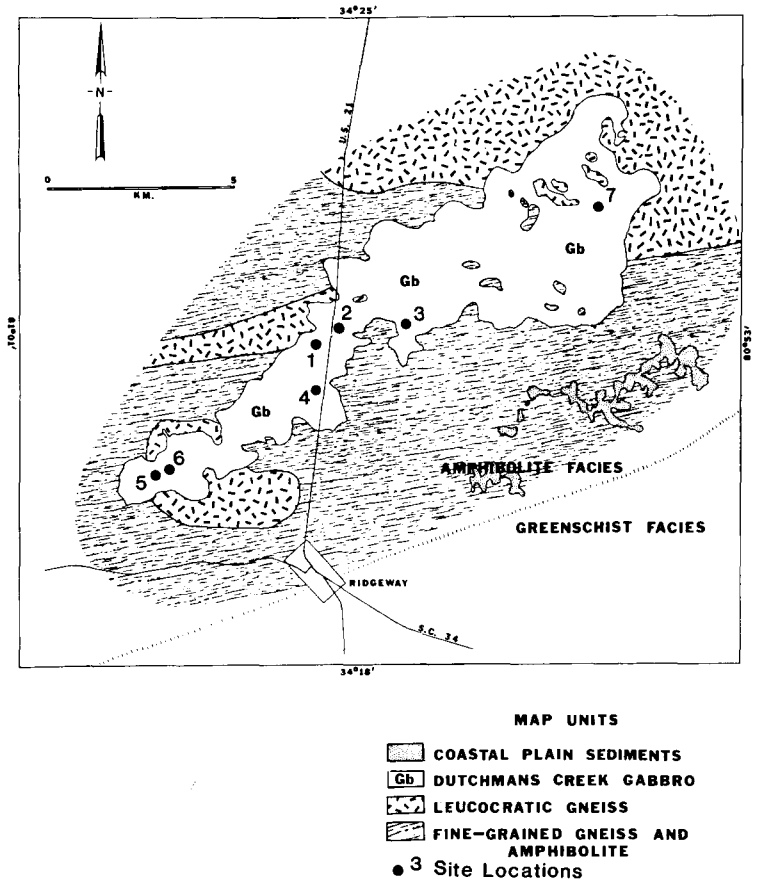


Fig. 7. Generalized geological map of the Dutchman's Creek pluton (After McSween and Nystrom, 1979) with sample localities.

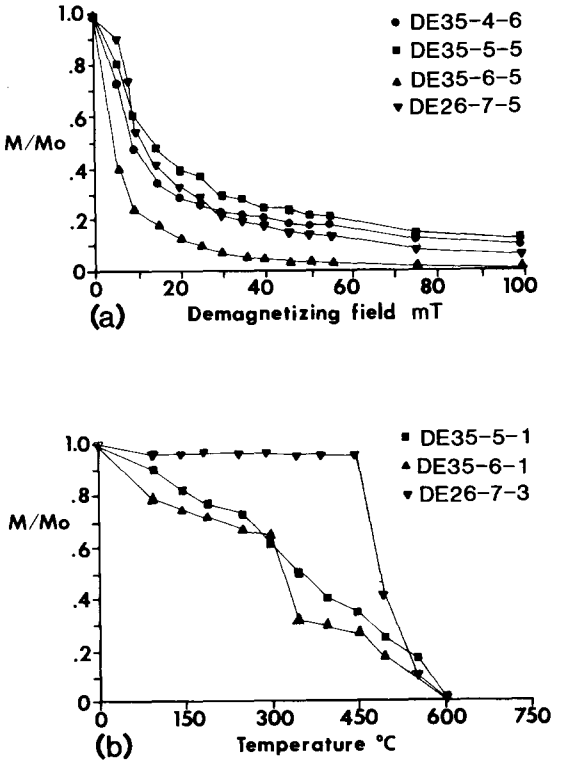


Fig. 8: Alternating frequency (a) and thermal (b) demagnetization spectra for the Dutchman's Creek pluton. Site 4 = DE35-4-5; site 5 = De35-5-5; site 6 = DE35-6-5; site 7 = DE26-7-5.

spectra than that for the Buffalo pluton. The NRM intensities vary from 2.2×10^{-2} A m^{-1} to 3.6×10^{-4} A m^{-1}, with an average of 4.4×10^{-3} A m^{-1}. Figure 10 shows cleaned, contrasted with NRM directions, for three of the seven sites in the Dutchman's Creek gabbro. Thermal demagnetization suggests that above 350°C, the characteristic magnetization is revealed (Fig. 8). The summary of the tilt corrected data appears in Table II with the calculated pole positions.

Like the Buffalo pluton, the Dutchman's Creek gabbro displays a single characteristic magnetic direction with a viscous component removed at low alternating fields. The best representation of the characteristic magnetization was found at 25 mT. The Curie temperature (Fig. 8) and the demagnetization data are consistent with titanomagnetite as the main carrier of remanence; however, some pilot samples display a thermal spectra typical of hematite. McSween and Nystrom (1979) have described hematite as an alteration product (above), but it is not apparent in all of the samples examined (Fig. 8). The absence of any indication of mineral grains with overlapping coercivity ranges suggests, that either both generations of magnetite and hematite formed close in time or, that the whole body has been remagnetized by a process which took it above its Curie point—a highly improbable situation, for which no evidence exists.

The mean value of remanence and the relevant Fisher statistics are shown in Table I and plotted in Fig. 11 with individual site means. The mean data for an additional 12 sites of the Dutchman's Creek gabbro (Zupan, personal communication, 1982) have been included in Table I. Thus, for this paper the best average declination and inclination is 161.3°, +27.7° with an $\alpha 95$ value of 6.0°.

Preliminary conventional K–Ar determinations of 396 ± 8 Ma and 385 ± 8 Ma for two whole rock specimens and 499 ± 10 Ma for a hornblende mineral separate from the Dutchman's Creek gabbro, have been done (Table I). The hornblende sample is particularly low in K (0.097%) and even small amounts of excess ^{40}Ar contamination would greatly alter the calculated age, thus the age of 499 ± 10 Ma may reflect the effect of excess ^{40}Ar contamination. The whole rock samples, on the other hand, are sufficiently high in K (0.600%) that small amounts of excess ^{40}Ar may have had only a small effect on the calculated age. While the 396 and 385 Ma dates could reflect the presence of small amounts of excess ^{40}Ar, the possibility of argon loss also exists, and these apparent ages must be regarded as the best approximation of the maximum age range of the Dutchman's Creek gabbro. $^{40}Ar/^{39}Ar$ studies are in progress to resolve this question.

2.3. The Jenkinsville diorites

Samples were obtained from sites in two diorites, which intrude amphibolitic country rock within the Charlotte belt, close to the contact with a migmatitic zone surrounding the Winnsboro granite (Fig. 12, Smith, 1982). Minor diorite intrusives are ubiquitous along the margin of the granite

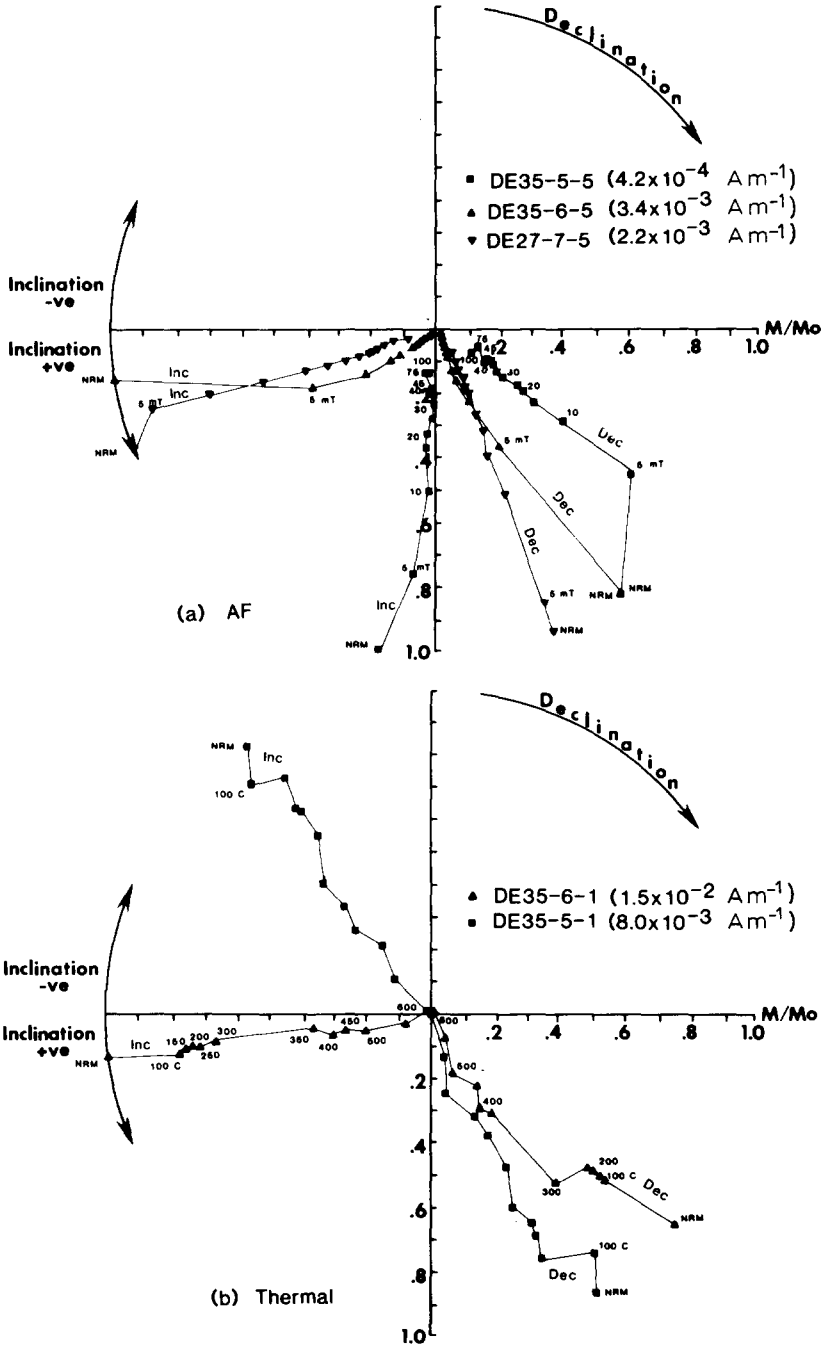


Fig. 9. Stable end point curves for alternating frequency (a) and thermal (b) demagnetization of Dutchman's Creek pilot samples. NRM intensity values are given in A m^{-1} (in parentheses).

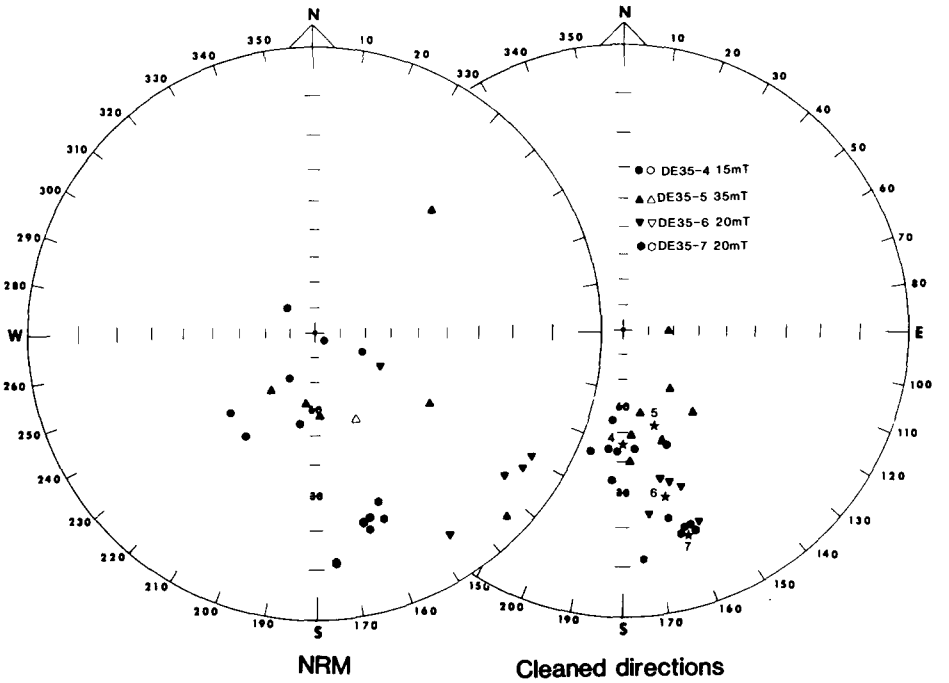


Fig. 10. NRM directions contrasted with the best cleaned directions for the Dutchman's Creek pluton. The stars represent site mean directions. Analytical sample numbers corresponding to site numbers are given in the caption for Fig. 8.

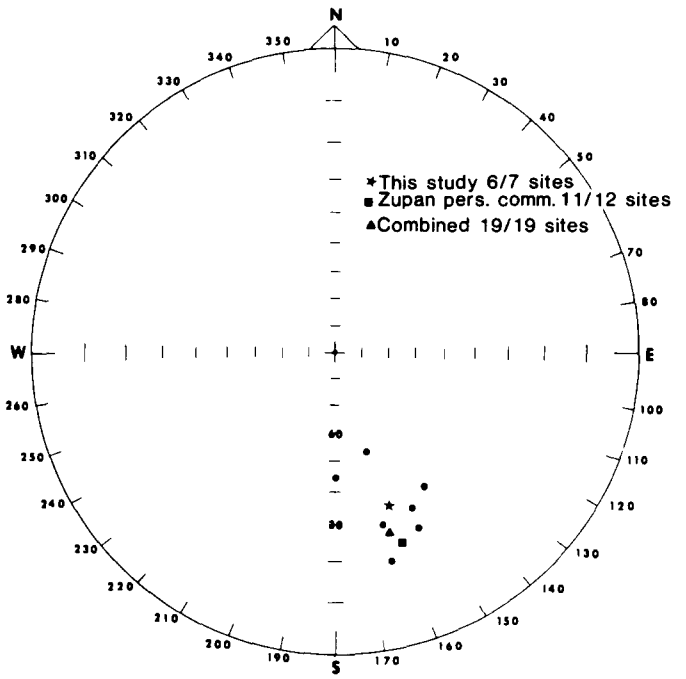


Fig. 11. Site mean directions for the 7 sites in this study. The star represents the overall mean directions for these sites. Zupan's overall mean direction for 11 of 12 sites is given by the square and the average of both studies is represented by the triangle for the 19 sites.

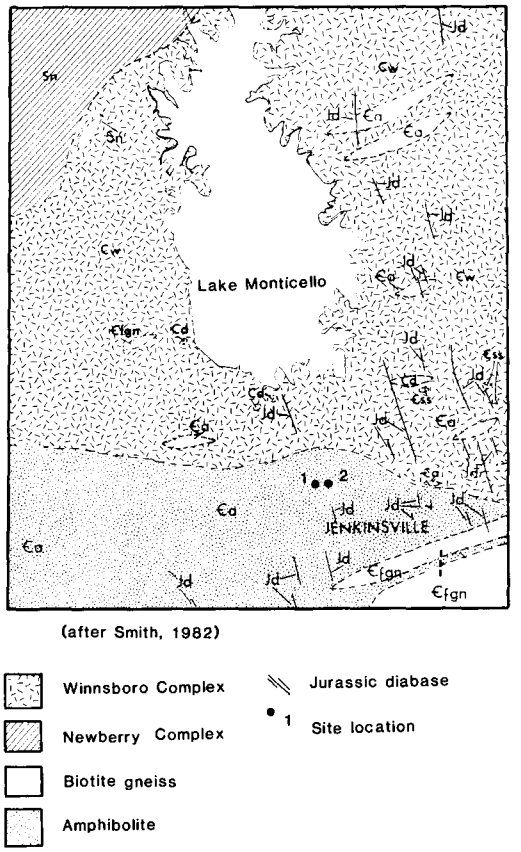


Fig. 12. Geological map of the Jenkinsville $7\frac{1}{2}$ minute quadrangle. Site 1 = CD46-1-6 and site 2 = CD46-2-2.

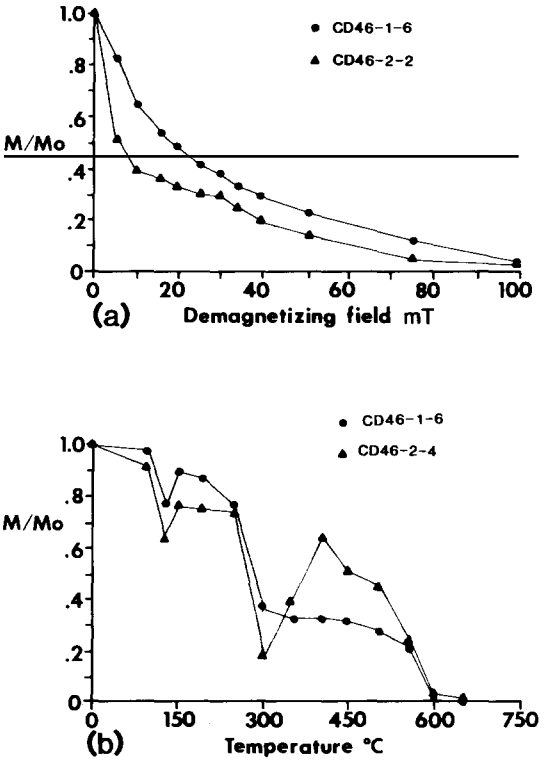


Fig. 13. Alternating frequency (a) and thermal (b) demagnetization spectra for the Jenkinsville diorites.

and throughout the southern Appalachian Piedmont. Their age is unknown, however, as they are post-metamorphic and intrude amphibolite–almandine facies metamorphic rocks of the Charlotte belt, they must be post Taconic (440–480 Ma old; Butler, 1972).

The results of magnetic cleaning of the samples measured are summarized in Table I. NRM intensities range from 7.7×10^{-4} A m^{-1} to 1.0×10^{-2} A m^{-1} and average 3.2×10^{-3} A m^{-1}. Normalized AF and thermal demagnetization curves are shown in Fig. 13, and stable end point curves are presented in Fig. 14. Thermal demagnetization seems to result in sudden jumps, both in the direction and intensity of magnetization of some samples. It is not clear whether the magnetization changes reflect errors in techniques, since most of the samples do not respond in this way, or whether they represent secondary components of magnetization. The vector removed at these stages is random in direction. Such samples were excluded from the mean computations. Figure 15 shows site means for the diorites.

Two single whole rock K–Ar apparent ages of 288 ± 6 Ma and 265 ± 5 Ma were determined for the diorites (Table I). It is not known if this age is the age of crystallization, or whether it represents the age of cooling due to uplift and erosion.

2.4. The Clouds Creek plutonic complex

The Clouds Creek igneous complex consists primarily of a massive to gneissic porphyritic granite pluton and a heterogeneous mafic complex developed on the eastern flank of the felsic portion of the complex (Secor and Snoke, 1978). The commonest rock in the mafic complex is a

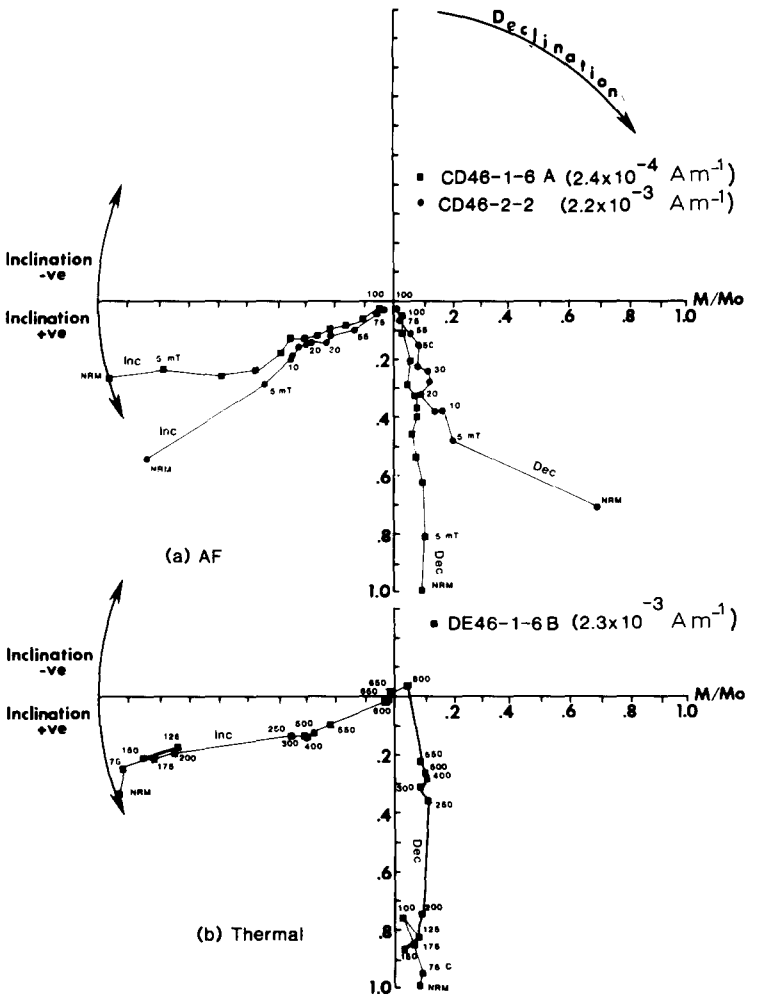


Fig. 14. Stable end point curves for alternating frequency (a) and thermal (b) demagnetization spectra for two pilots of the Jenkinsville diorites. NRM intensities are given in A m^{-1} (in parentheses).

biotite–hornblende quartz diorite and minor amounts of gabbro of variable texture and modal composition. Serpentinization has occurred irregularly throughout the mafic complex. The Clouds Creek granite is located within the Slate belt and bounded on its southern edge by the Modoc fault zone and the Kiokee belt (Fig. 16) near Batesburg, South Carolina. While the mafic and felsic phases of the Clouds Creek complex seem broadly contemporaneous, their age relationship is not clear. Granitic cupolas penetrate the dioritic pluton, indicating an older age for the mafic body (Secor and Snoke, 1978). The mafic rocks have also been subjected to a rather intense static alteration, similar to the retrograde nature of the contact aureole that partially surrounds the felsic portions of the complex, which further supports an older age for the mafic portion (Secor and Snoke, 1978).

Paleomagnetic techniques were the same for the granitic and mafic rocks as in previous sections, and Table I provides site mean directions for five sites within the mafic complex and one site within the granite. The remanence of samples from this one site in the granite is reported, even though its

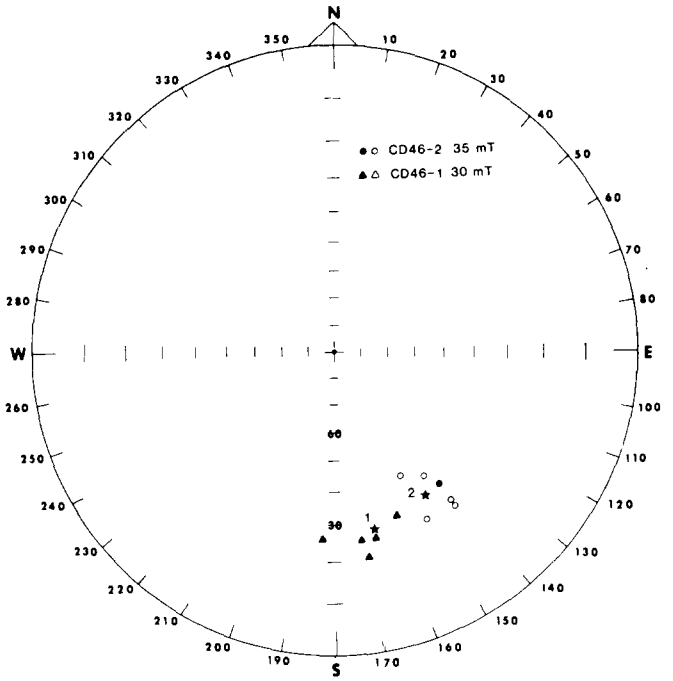


Fig. 15. Site mean directions for two diorite sites in the Jenkinsville quadrangle. The stars represent sites 1 and 2. The best cleaned directions of the two sites are 30 and 35 mT.

α95 value is large, for comparison with the data from samples of the basic rocks.

2.5. *The mafic complex*

AF and thermal demagnetization spectra of the mafic samples (Fig. 17) show the same rapid decrease in magnetic intensity, after treatment in relatively low alternating fields, as found in the Buffalo, Dutchman's Creek and Jenkinsville plutons. Stable end point curves for AF and thermal demagnetization (Fig. 18) show similar behavior, with a viscous magnetic direction removed around 5 mT and the remanence retrieved between 20 and 25 mT for four of the sites. Site 5 did not respond to magnetic cleaning and was excluded from the calculation.

The site mean direction for four of five sites (Table I) is 153.4° declination, 6.8° inclination with an α95 value of 15.6°, corresponding to a pole position of 45.4°N 137.9°E (Figs. 19, 20) and Table II shows tilt corrected values for the pole position.

2.6. *The granitic complex*

AF and thermal demagnetization spectra were produced for several granite pilot samples from the Clouds Creek complex (Fig. 21). It is interesting to note that while a stable remanence could not be retrieved from most of the samples, two antipodal directions were obtained (Fig. 22). The first direction is similar to that retrieved for the gabbros (for example, the Buffalo gabbro), with declinations in the vicinity of 160° and inclinations between 10 and 20°. The second direction seen in the demagnetization curves of the granites is antipodal with declination and inclination values near 340°, −20°, respectively (Fig. 23). It is possible that cooling of the Clouds Creek complex occurred over a sufficiently long period of time for the felsic complex to record a magnetic reversal, but the occurrence of both directions at the same site suggest that this magnetization is secondary. The results of Brown and Barton (1980) and Barton and Brown (1982), for the Churchland granite

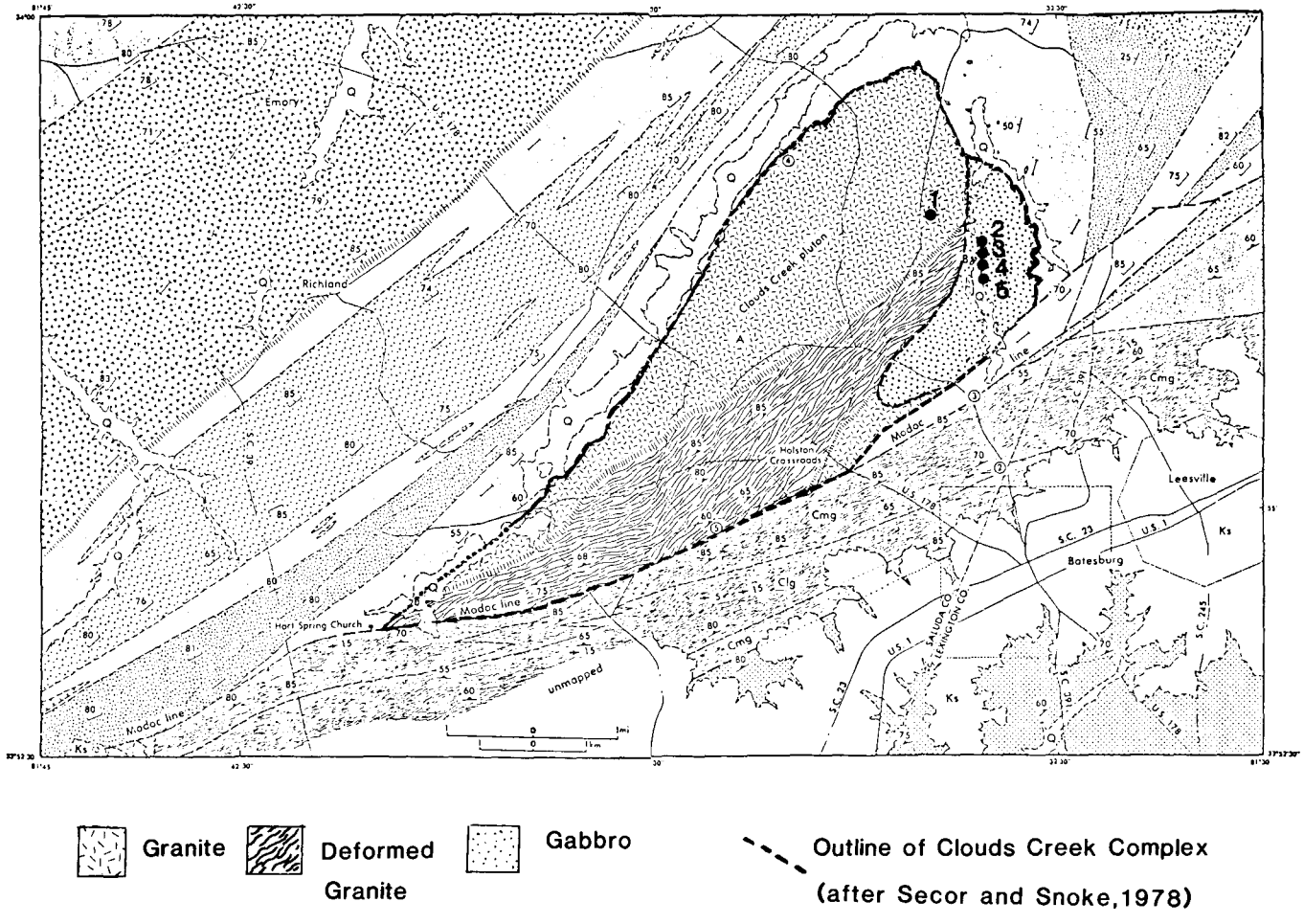


Fig. 16. Generalized geological map of the Batesburg and Emory $7\frac{1}{2}$ minute quadrangles near the Batesburg area with sample localities in the Clouds Creek igneous complex.

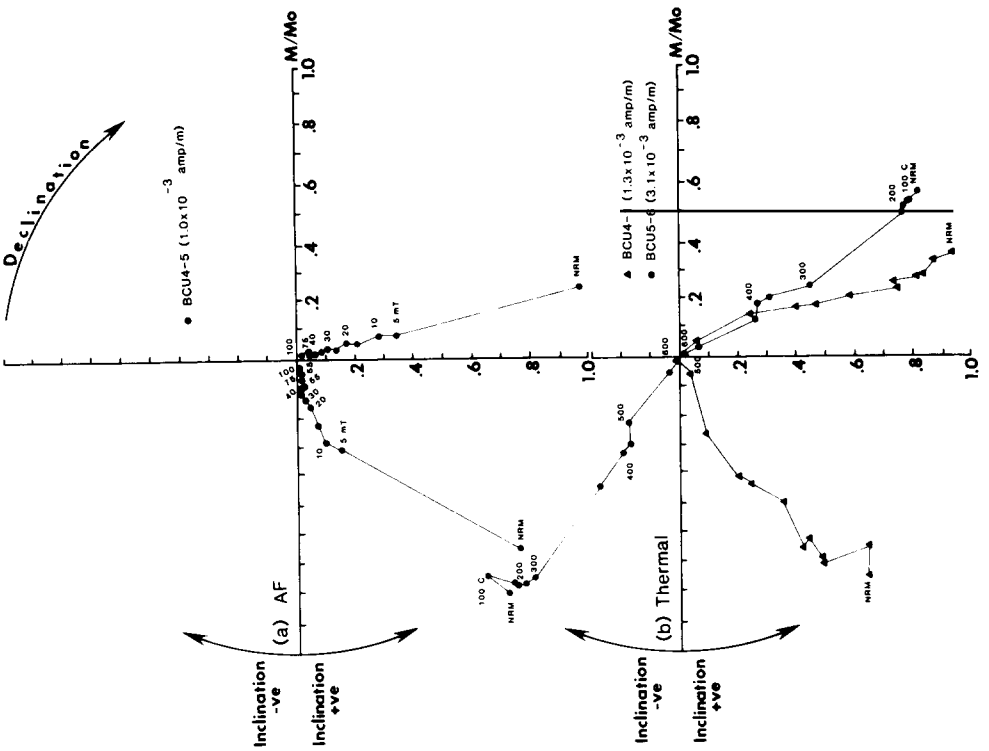


Fig. 18. Stable end point curves for alternating frequency (a) and thermal (b) demagnetization spectra for pilots in the Clouds Creek gabbro. NRM intensities given in A m^{-1} (parentheses).

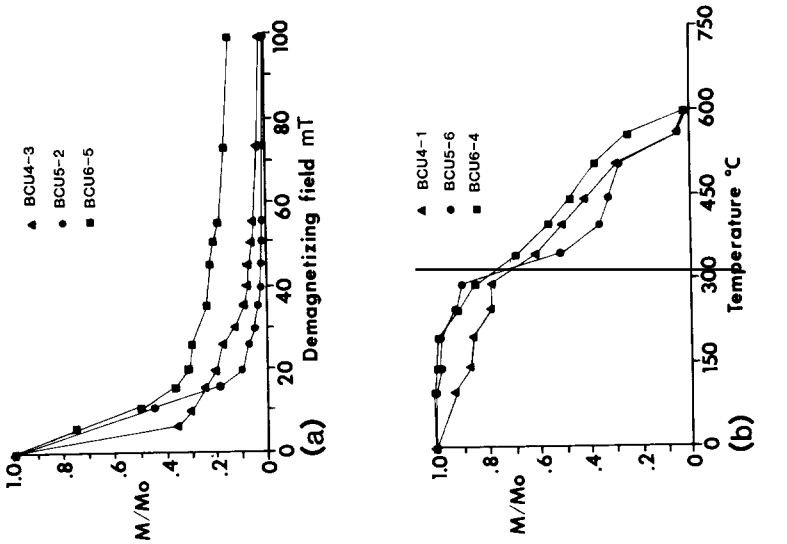


Fig. 17. Alternating frequency (a) and thermal (b) demagnetization spectra of the Clouds Creek gabbro. Site 3 = BCCU-4; site 4 = BCCU-5; site 5 = BCCU-6.

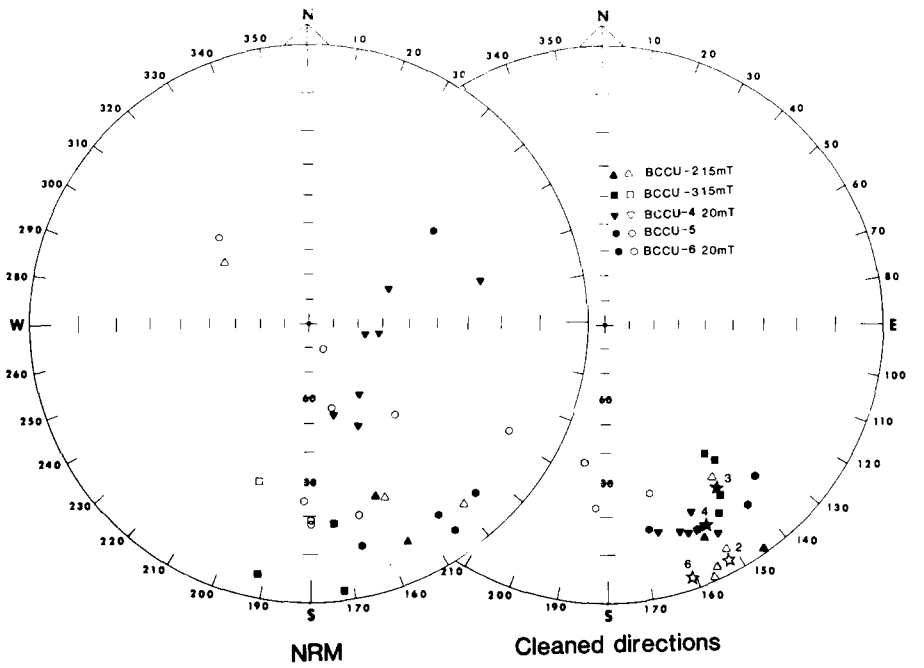


Fig. 19. NRM directions contrasted with the best cleaned directions for the Clouds Creek gabbro. The stars represent site mean directions for four sites. Analytical sample numbers corresponding to site numbers are given in the caption of Fig. 17.

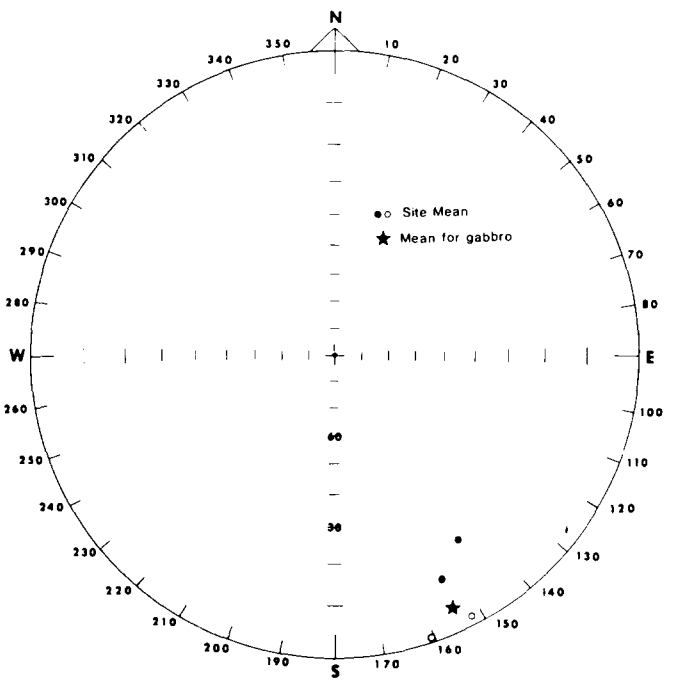


Fig. 20. Site mean directions for four sites with the overall mean for the Clouds Creek gabbro represented by the star.

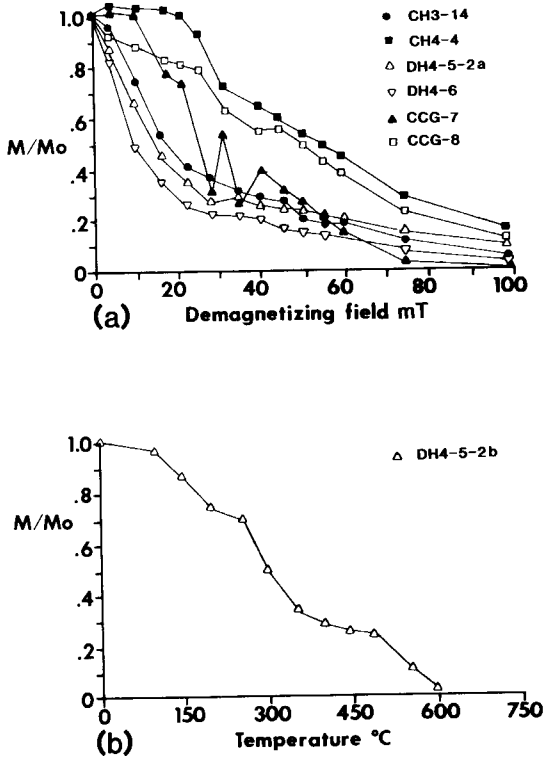


Fig. 21. Alternating frequency (a) and thermal (b) demagnetization spectra for pilot samples from the Clouds Creek granite.

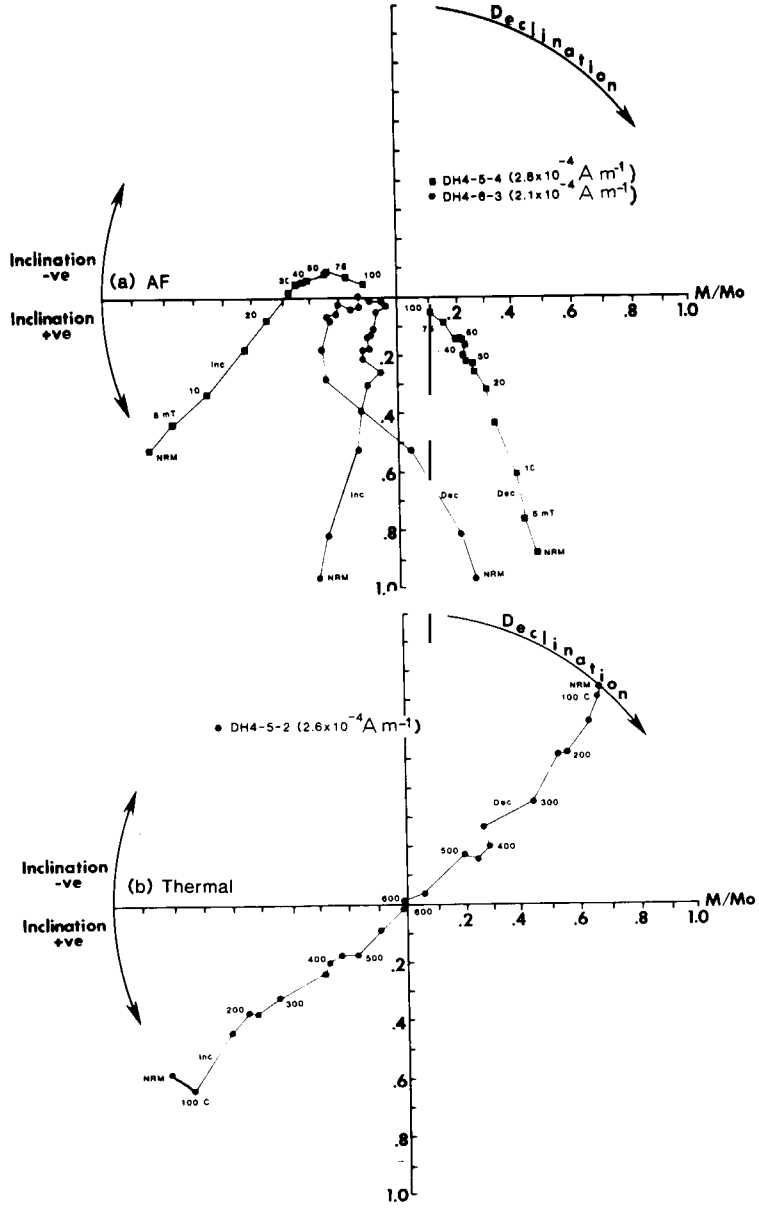


Fig. 22. Stable end point curves for alternating frequency (a) and thermal (b) demagnetization spectra for the Clouds Creek granite. NRM intensities are given in A m^{-1} (parentheses).

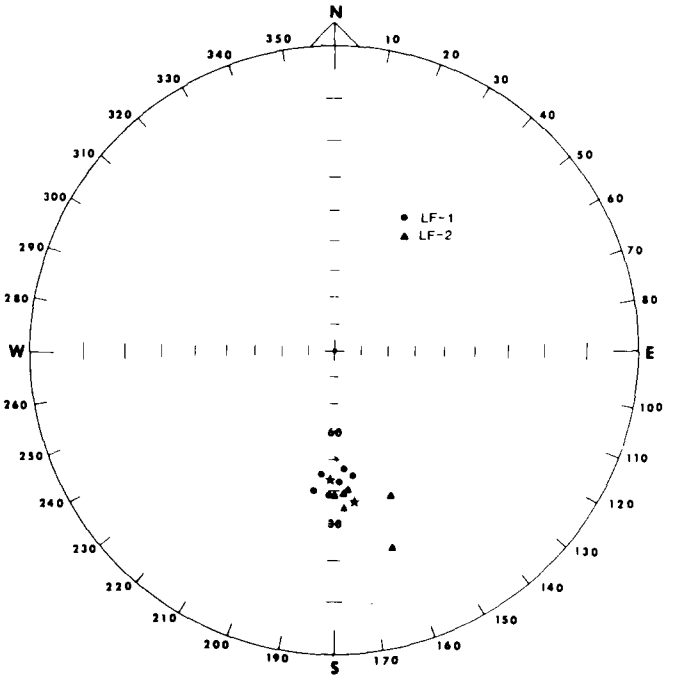


Fig. 23. Site mean directions for two sites in the Little Fork gabbro within the Pageland granitic igneous complex (Zupan, personal communication, 1982). The stars give site mean directions.

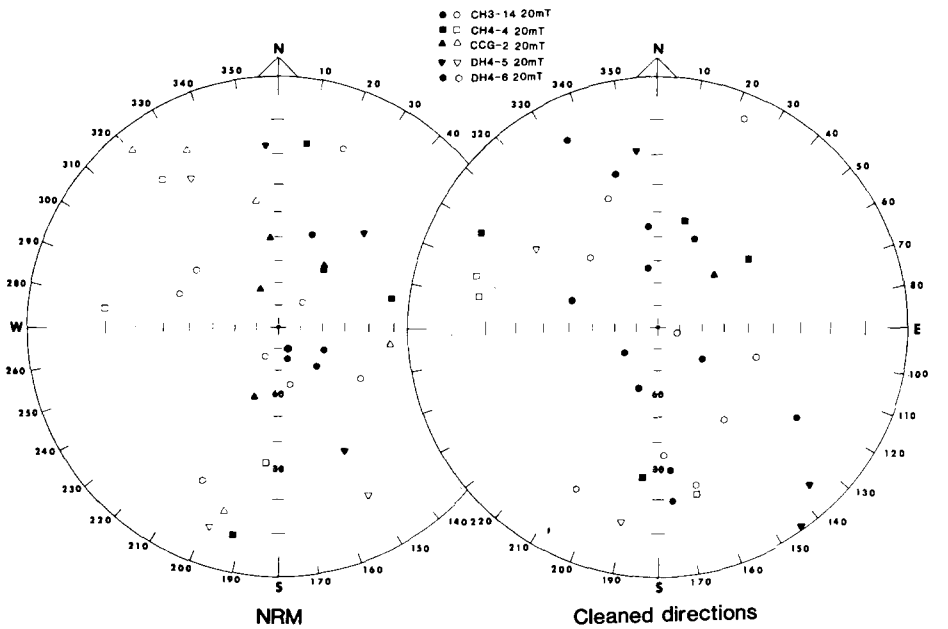


Fig. 24. NRM directions contrasted with the best cleaned directions for several Clouds Creek granite sites. Mean directions were not calculated because of the large degree of dispersion within each site.

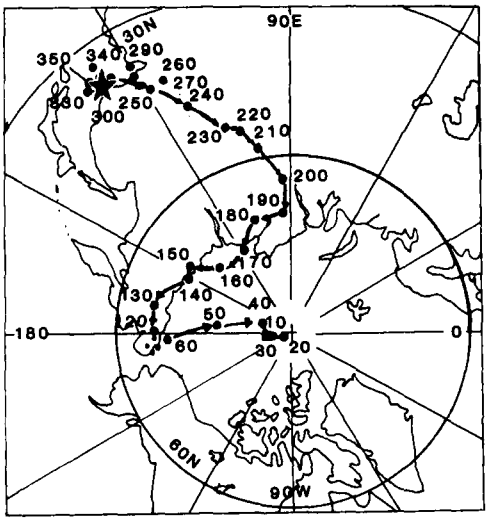


Fig. 25. Apparent polar wander path (APWP) relative to North America since the Devonian (after Irving, 1977, 1979) with the mean pole position of this study plotted (star).

(341° dec., −24.7° inc.) and the Lilesville granite (334.4° dec., −16.3° inc.), and tentative results by Zupan (personal communication, 1982) for the Pageland granite, all record reversed directions when compared to the mafic rocks described here. Furthermore, the Little Fork gabbro also yields antipodal directions to the Pageland granite into which it intrudes (Fig. 24).

Rb–Sr whole rock isochron ages of 319 ± 27 Ma and a biotite whole rock date of 313 ± 2 Ma have been provided by Fullagar and Butler (1979) for the Clouds Creek granite. There are no radiometric ages for the mafic portion of the complex. However, a Pennsylvanian age is inferred from radiometric evidence for at least the granitic portion of the pluton (Fullagar, 1971; Fullagar and Butler, 1979; Fullagar and Kish, 1980; Fullagar, 1981).

3. Discussion

All of the gabbroic bodies in this study display a single characteristic magnetization. The magnetization consists of a soft component, largely removed by 5 mT and a stable hard magnetic component usually isolated between 15 to 25 mT. The magnetic behavior of the Buffalo, Dutchman's Creek, Clouds Creek and Little Fork gabbros suggest that titanomagnetite is the principal carrier of the remanence, though there are a few cases where hematite is the carrier. The magnetization of these rocks appears to reflect a relatively simple history of crystallization and cooling, and there is nothing observable in their petrology, as seen in thin section, that is inconsistent with this interpretation.

The conventional whole rock K–Ar apparent ages, listed in Table I are consistent with what might be anticipated from their relationships to dated granite bodies (Fullagar, 1971; Fullagar and Butler, 1979; Fullagar, 1981). The one exception to this is the old date obtained for samples of the Buffalo pluton which yielded ages in excess of 1 Ga. This is interpreted as being due to the presence of excess ^{40}Ar, demonstrated as a significant factor affecting the ages of mafic bodies in the southern Appalachians by Dooley and Wampler (1983). $^{40}Ar/^{39}Ar$ data is necessary to further refine the interpretation of their age.

The four gabbro bodies and two diorite bodies in this study have declinations and inclinations similar to each other and thus are most likely to be broadly contemporaneous. These values correspond to an overall average pole position of 38.9°N 120.8°E, with an $\alpha 95$ value of 10.2° (Table I). This pole position plots in the vicinity of the 300 Ma age point of Irving's (1977, 1979) apparent polar wander path for North America (Fig. 25). However, the $\alpha 95$ value of 10.2° encompasses pole positions in the range of 250–360 Ma. Thus, within experimental error, our pole position is not statistically different from those determined for rocks in the age range of 250–360 Ma. One of these gabbro bodies (the Clouds Creek gabbro) has a declination of 160° and inclination of +30° and is in intrusive contact with the Clouds Creek granite, an igneous intrusion with a Rb–Sr age of 319 Ma (Fullagar, 1971, 1981). Thus, provided the gabbro is only slightly different in age from the granite (Secor and Snoke, 1978), the age of the granite offers evidence suggesting an age near 319 Ma for the gabbros of this paper.

The 10° westerly tilt correction about the Appalachian trend, suggested by Dooley and Smith

(1982) on the basis of their study of late Triassic–early Jurassic diabase dikes of South Carolina, when applied to the gabbro data, results in a pole position closer to a 350 Ma pole position (Irving, 1977; 1979). The test for tilt correction is therefore inconclusive, but might be more meaningful if the radiometric ages could be better defined (Table II, Fig. 25).

The simplest interpretation of the data from the mafic rocks is that they reflect the directions of the geomagnetic field of late Devonian–Carboniferous time. The proximity of the geomagnetic poles to appropriate poles from stable cratonic North America leaves little room for displaced terrains, though with the precision of the pole positions no conclusive statements can be made for horizontal displacements of the order suggested by COCORP data. These data therefore augment and complement the data from the granitic rocks (Brown and Barton, 1980; Ellwood, 1982; Barton and Brown, 1982) and suggest that if displaced terrains exist, evidence must be sought in studies of older rocks, or in different geologic belts.

The directions of remanence of the gabbros in Table I are of reversed polarity. Brown and Barton (1980) and Ellwood (1982) record reversed polarity in other intrusive bodies of similar age. This is consistent with intrusion during the Kiaman interval, the base of which lies at 300 Ma (Irving, 1964).

4. Conclusions

In this study the gabbros with single characteristic directions of magnetization were examined. Their calculated pole positions agreed well enough to allow a single pole position computation to be made for the whole group, yielding a paleomagnetic pole position of 38.9°N 120.8°E. This position is close to the 300 Ma old position on the APWP of North America and on the basis of polarity, consistent with intrusion during the Kiaman interval. This pole position has a precision ($\alpha 95$) of 10.2° and encompasses positions on the APWP between 250 and 360 Ma. Tentative radiometric age determinations by conventional K–Ar techniques show a bimodal distribution of apparent ages at 360–390 Ma and at 1.0–1.6 Ga. The older group is clearly anomalous, because gabbros with these ages are older than the country rocks they intrude. The ages of the younger 360–390 Ma group are more difficult to interpret. If small amounts of excess ^{40}Ar were present then the ages would represent a maximum age for the gabbros. $^{40}Ar/^{39}Ar$ techniques will resolve some of the problems in interpreting the conventional K–Ar apparent ages for gabbros with single characteristic magnetic directions. Thus a conservative estimate of the age of the gabbros is that they must lie within 300–360 Ma age range on the basis of the present radiometric and paleomagnetic data.

Applying a 10° westerly tilt, about an Appalachian trend, places the pole position of this study more nearly into coincidence with a 350 Ma old position on the APWP. Thus the tilt correction is inconclusive and awaits further resolution of radiometric dating.

There is little evidence for displaced terrains in the southern Appalachians for rocks of the age of the gabbros in this paper. Evidence for displaced terrains must be sought for in older rocks, or in other geologic belts. The 10.2° circle of confidence cannot distinguish small rotations about a vertical axis, as would be required for low angle thrusting of the type and degree (200 km) suggested by COCORP data (Cook et al., 1979), or of the degree suggested (50 km) by Iverson and Smithson (1982). The interpretation of Iverson and Smithson (1982) is perhaps most plausible in light of this paleomagnetic data and recently published data (Ellwood, 1982), because no great tectonic movements would be required for such an interpretation, a fact that appears to agree with present paleomagnetic data.

Acknowledgments

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